

Color Sorting Machine with Conveyor Belt

1. Overview

This project is an automatic color sorting machine. Objects (such as colored balls, beads or blocks) are carried on a belt driven by a geared DC motor. An IR sensor detects each object as it arrives, the TCS34725 sensor reads its color, and the ESP32 turns one of two servo motors to guide the object into the correct sorting box. A 1.3" OLED display shows the live count of each color with its name.

Objectives

- Detect objects automatically and identify their color.
- Sort objects into separate boxes using two servo gates.
- Show the name and count of each color on the OLED in real time.
- Complete one sorting cycle per object in 1.5 seconds, with a 1 second interval before the next object.
- Run from a battery with a single on/off switch.

2. Components

Component	Qty	Purpose
ESP32 development board	1	Main controller; reads sensors, drives motor and servos, updates display
L298N motor driver	1	Controls speed and direction of the belt motor.
Reduction (gear) DC motor	1	Drives the conveyor belt at low speed with good torque
GT2 timing pulley, 20 teeth, 8 mm bore (for 6 mm belt)	2	Mounted on motor shaft and roller shaft to transfer the drive through the timing belt
GT2 122mm Closed Loop Timing Belt 6mm Width	1	Transfers motor drive to the conveyor roller
Conveyor belt, 65 mm wide × 760 mm long	1	Carries the objects along the sorting path
Roller, 78 mm length Diameter 34mm	2	Drive roller and idler roller that hold and move the conveyor belt
Steel rod, 5 mm diameter, 100 mm length	1	Shaft for a roller
Steel rod, 5 mm diameter, 120 mm length	1	Shaft for a roller (extra length for the pulley)
Bearing: inner diameter (d) 5 mm, outer diameter (D) 13 mm, width (B) 4 mm	4	Supports the roller rods so they rotate smoothly
IR sensor module	1	Detects that an object is in front of the color sensor
TCS34725 color sensor	1	Measures red, green, blue and clear light values over I2C
OLED display 1.3"	1	Shows color names and counts
Servo motor MG996R	2	Move the gates that guide objects into the boxes
On/off switch	1	Main power switch
Battery pack	1	Power source (for example 7.4 V Li-ion,)
3D model (3D printed frame and parts)	1 set	Holds the conveyor, sensors, gates and sorting boxes
Jumper wires, sorting boxes	-	Connections and collecting the sorted objects

3. Working principle

1. Switch on. The ESP32 starts the OLED, color sensor, servos (both at home position 0°) and the belt motor. The OLED shows the three counts, all zero.
2. The belt motor runs continuously at PWM speed 150 and the objects move forward.
3. When an object cuts the IR beam, an interrupt (falling edge on GPIO 33) fires. It brakes the motor instantly by shorting the motor terminals, then the main loop holds the brake for 80 ms and cuts the enable.
4. The system enters color sensing state for up to 1 second. The TCS34725 reads R, G, B and Clear values repeatedly until a color matches.
5. When a color is found, the belt restarts and the OLED shows the color name in large text at the center of the screen.
6. The matching servo turns to 60° to guide the object into its box, then returns to 0°. The count of that color increases by one.
7. The OLED returns to the count screen and the machine waits for the next object.
8. If no color is recognised within 1 second (dark, white or unknown object), the belt restarts and nothing is counted.

Sorting logic with two servos

Detected color	Action	Servo hold time in code	Result
Orange	Servo 1 turns to 60°	1.0 s + 0.3 s return	Falls in Box 1
Green	Servo 2 turns to 60°	1.5 s + 0.3 s return	Falls in Box 2
Blue	No servo	0.5 s delay	Passes straight to Box 3 (end of belt)
White / dark / unknown	Ignored, not counted	-	Passes straight through

Two servos sort three colors: two gates for orange and green, and the end of the belt for blue.

Sorting cycle time and interval

Each object takes **1.5 seconds** from detection to the count being updated. After that the machine waits **1 second** before the next object, so one object is handled every 2.5 seconds (about 24 objects per minute).

Stage	Time (approx.)
IR detects object and belt stops	0.2 s
Object settles under the sensor	0.2 s
TCS34725 reads and ESP32 decides the color	0.2 s
Servo gate moves to the sorting position	0.3 s
Belt restarts and object drops into the box	0.5 s
Gate returns to neutral and OLED count updates	0.1 s
Sorting cycle time per object	1.5 s
Interval for the next object	1 s
Total time per object	2.5 s

The stage split is an approximate breakdown; adjust it to match your measured timings.

4. Circuit and wiring

Power

- Battery positive goes through the switch to the **12V / Vin** terminal of the L298N. Battery negative goes to **GND** of the L298N.
- With the 5V-enable jumper fitted (supply up to 12 V), the L298N **5V** output powers the ESP32 **VIN** pin and both servos.
- All grounds (battery, L298N, ESP32, servos, sensors) must be joined together.
- Add a 470 μ F capacitor across the 5 V rail near the servos to avoid resets when they move.

Pin connections

Device	Device pin	ESP32 pin
TCS34725	VIN	3V3
	GND	GND
	SDA	GPIO 21
	SCL	GPIO 22
OLED 1.3" I2C	VCC	3V3
	GND	GND
	SDA	GPIO 21 (shared)
	SCL	GPIO 22 (shared)
IR module	VCC	3V3
	GND	GND
	OUT	GPIO 33 (interrupt, falling edge)
L298N	ENA (remove jumper)	GPIO 25 (PWM speed, value 150)
	IN1	GPIO 26
	IN2	GPIO 27
Servo 1 (orange)	Signal	GPIO 18
Servo 2 (green)	Signal	GPIO 19
Motor	Two wires	L298N OUT1 and OUT2

The TCS34725 (address 0x29) and the OLED (address usually 0x3C) share the same I2C bus because their addresses are different. The 1.3" OLED normally uses the SH1106 controller, not SSD1306, so select the correct driver in the library.

5. Mechanical design

- **Conveyor belt:** a 65 mm wide, 760 mm long belt runs over two 78 mm rollers. Each roller turns on a 5 mm rod supported by $5 \times 13 \times 4$ mm bearings (rod lengths 100 mm and 120 mm).
- **Belt drive:** one GT2 20-teeth pulley (8 mm bore, for 6 mm belt) is fixed on the gear motor shaft and the second on the drive roller shaft. The timing belt runs between them. Keep it tight but not overstretched.
- **Frame:** the 3D model holds the rollers, sensors, servo gates and boxes in position.
- **Sensor position:** fix the TCS34725 about 5–10 mm above the object path, facing down. Place the IR module just before it so the object stops under the color sensor.
- **Lighting:** shield the sensor from room light with a small cover. Use the sensor's own LED (on) for steady readings.
- **Gates:** attach a small paddle to each servo horn. In neutral it lies along the path; when opened it deflects the object into the box.
- **Boxes:** place one box under each gate and a third at the end of the belt.

6. Software

Libraries (Arduino IDE)

- Adafruit_TCS34725 - color sensor
- U8g2 - 1.3" OLED (SH1106 driver)
- ESP32Servo - servo control on ESP32

Program flow

```
SETUP
  init serial, IR pin (pull-up), motor pins, servos (home 0 deg)
  init I2C (SDA 21, SCL 22), OLED, TCS34725
  show counts, attach IR interrupt (FALLING), start motor

LOOP
  state MOTOR_RUNNING:
    if object detected (set by IR interrupt):
      brake motor (80 ms), start 1 s timer
      state = COLOR_SENSING
    else keep motor running

  state COLOR_SENSING:
    read color
    if color found:
      restart motor, show color name on OLED
      orange -> servo 1 | green -> servo 2 | blue -> 0.5 s wait
      show counts, state = MOTOR_RUNNING
    else if 1 s timeout:
      restart motor, state = MOTOR_RUNNING
```

Color decision rules

Check (in order)	Condition	Result
Too dark	Clear < 20	Ignored
White	Clear ≥ 180 and (max - min of R, G, B) ≤ 8	Ignored
Orange	$R > G + 3$ and $R > B + 3$	Servo 1
Green	$G > R + 5$ and $G > B + 5$	Servo 2
Blue	$B > R + 5$ and $B > G + 5$	No servo

Orange has both red and green in it, so its dominance value is lower (3). The sensor uses 50 ms integration time and 4x gain.

OLED screens

Count screen	Detection screen
COLOR SORTING	(large, centered)
ORANGE: 12	ORANGE
GREEN : 08	
BLUE : 05	

7. Calibration

1. Open the Serial Monitor at 115200 baud. The code prints R, G, B and C values for every reading.
2. Place each sample object (orange, green, blue) under the sensor several times and note the values.
3. Tune `ORANGE_DOMINANCE`, `GREEN_DOMINANCE` and `BLUE_DOMINANCE` until each color is detected correctly.
4. Adjust `MIN_CLEAR_LEVEL`, `WHITE_CLEAR_LEVEL` and `WHITE_DIFFERENCE` if white or dark objects are wrongly counted.
5. Adjust `SERV01_SORT`, `SERV02_SORT` and the servo delay times so objects fall cleanly into the boxes.
6. Change `MOTOR_SPEED` (0-255) so the belt is slow enough for reliable detection.

8. Troubleshooting

Problem	Likely cause	Fix
ESP32 resets when servo moves	Voltage drop on 5 V rail	Add capacitor, use fresh battery, power servos separately with common GND
OLED blank	Wrong driver or address	Use SH1106 driver; run an I2C scanner
Color not detected / timeout	Thresholds too strict or object too far	Check Serial values and lower the dominance numbers
Wrong colors	Ambient light or distance	Shield the sensor, fix the height, recalibrate
Object double counted	IR still triggered	Wait until the IR clears before the next detection
Motor does not turn	ENA jumper left on or no PWM	Check ENA wiring and IN1/IN2 states
Belt slips	Loose belt or pulley screw	Tighten belt and grub screws

9. Applications and future improvements

- Sorting fruits, candies, recycled items or factory parts by color.
- Add more colors with extra servos or a rotating chute.
- Show counts on a web page over ESP32 Wi-Fi.
- Store totals in memory so they survive a restart.

10. Conclusion

The ESP32-based color sorter combines a belt conveyor, an IR trigger, a TCS34725 color sensor and two servo gates to sort objects automatically, while the OLED gives a clear live count for each color. The design is low-cost, battery powered and easy to extend.